

AGRITECH · ENGINEERING CONCEPT

SoilSense

IoT-Based Smart Soil Monitoring & Decision Support System

MEASURE · INTERPRET · INFORM

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01 · THE DECISION GAP

Soil changes faster than decisions do.

The challenge is not a lack of readings. It is the distance between a reading and a useful next step.

FIELD

Conditions need continuous monitoring.

WATER

Irrigation depends on moisture and water availability.

CROP CARE

Crop-care decisions depend on nutrient and pH conditions.

TRANSLATION

Raw readings are difficult to turn into action.



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02 · THE CORE IDEA

SoilSense turns readings into decisions.

A COMPACT LOOP FROM MEASUREMENT TO PRACTICAL GUIDANCE.

MEASURE → INTERPRET → INFORM



01

MEASURE

Collect NPK, pH, and moisture data.



02

INTERPRET

Microcontroller captures readings and applies interpretation logic.



03

INFORM

Surface soil status, alerts, and action-oriented guidance.

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03 · SYSTEM ARCHITECTURE

Three sensors. One clear signal.

A PROPOSED SENSING-TO-USER ARCHITECTURE FOR TURNING SOIL CONDITIONS INTO UNDERSTANDABLE INFORMATION.

PHYSICAL WORLD → SENSING → PROCESSING → ANALYSIS → USER → DECISION



PHYSICAL WORLD →

SOIL / FIELD CONDITIONS

Moisture · nutrients · pH



SENSING →

NPK · pH · MOISTURE SENSORS



PROCESSING →

MICROCONTROLLER

Data processing



ANALYSIS →

SOIL-CONDITION ANALYSIS

Soil status



USER →

DASHBOARD / ALERT

Farmer-facing information



DECISION

DECISION SUPPORT

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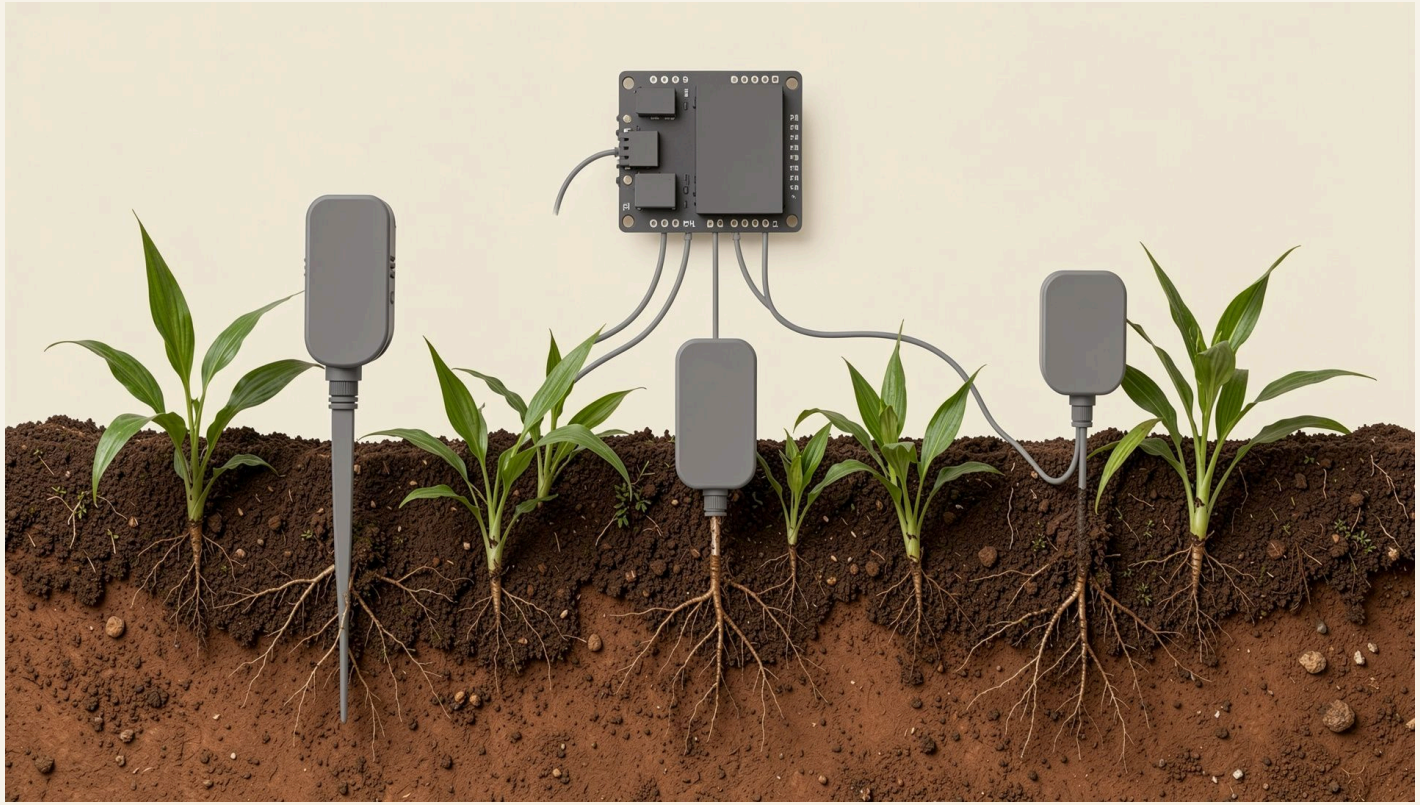
04 · INSIDE SOILSENSE

From the field to the microcontroller.

PROPOSED HARDWARE CONFIGURATION — A CLEAR STARTING POINT FOR IMPLEMENTATION.

PROPOSED HARDWARE CONFIGURATION

NPK · PH · MOISTURE SENSING IN A CONNECTED SOIL/CROP ENVIRONMENT.



NPK SENSOR

Soil nutrient sensing

PH SENSOR

Soil acidity sensing

SOIL MOISTURE SENSOR

Water availability sensing

MICROCONTROLLER

Receives sensor signals

THE ARCHITECTURE IS PRESENTED AS A PROPOSED CONFIGURATION, NOT A CLAIM OF COMPLETED FIELD HARDWARE.

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05 · DATA FLOW

Every reading has a purpose.

THE SYSTEM MOVES FROM PHYSICAL CONDITIONS TO INFORMATION, THEN TO ACTION.

RAW DATA → PROCESSING → MEANING → ACTION



01

PHYSICAL WORLD

Moisture · nutrients · pH



02

CAPTURE

Sensors
Raw readings



03

PROCESS

Microcontroller
Data processing



04

INTERPRET

Soil condition analysis
Soil status



05

INFORM

Action-oriented information



06

ACT

Farmer decision

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06 · INNOVATION & DIFFERENTIATION

The difference is not sensing. It is understanding.

ILLUSTRATIVE COMPARISON OF THE INTERPRETATION GAP.



TRADITIONAL SENSOR APPROACH

Sensor → raw reading

Uncontextualized number

Farmer interprets the number.

CONFUSING RAW-NUMBER TREATMENT



SOILSENSE

Sensor → reading

→ Interpretation → soil status

ILLUSTRATIVE EXAMPLE

SOIL STATUS
Moisture: Adequate
pH: Within range
Nutrients: Requires attention

ACTION-ORIENTED INFORMATION

SoilSense adds a meaning layer between measurement and decision.

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07 · EXPECTED IMPACT

Better information at the moment it matters.

The value proposition is decision support — not a promise of quantified outcomes.

Understand

Support better understanding of soil conditions.

Decide

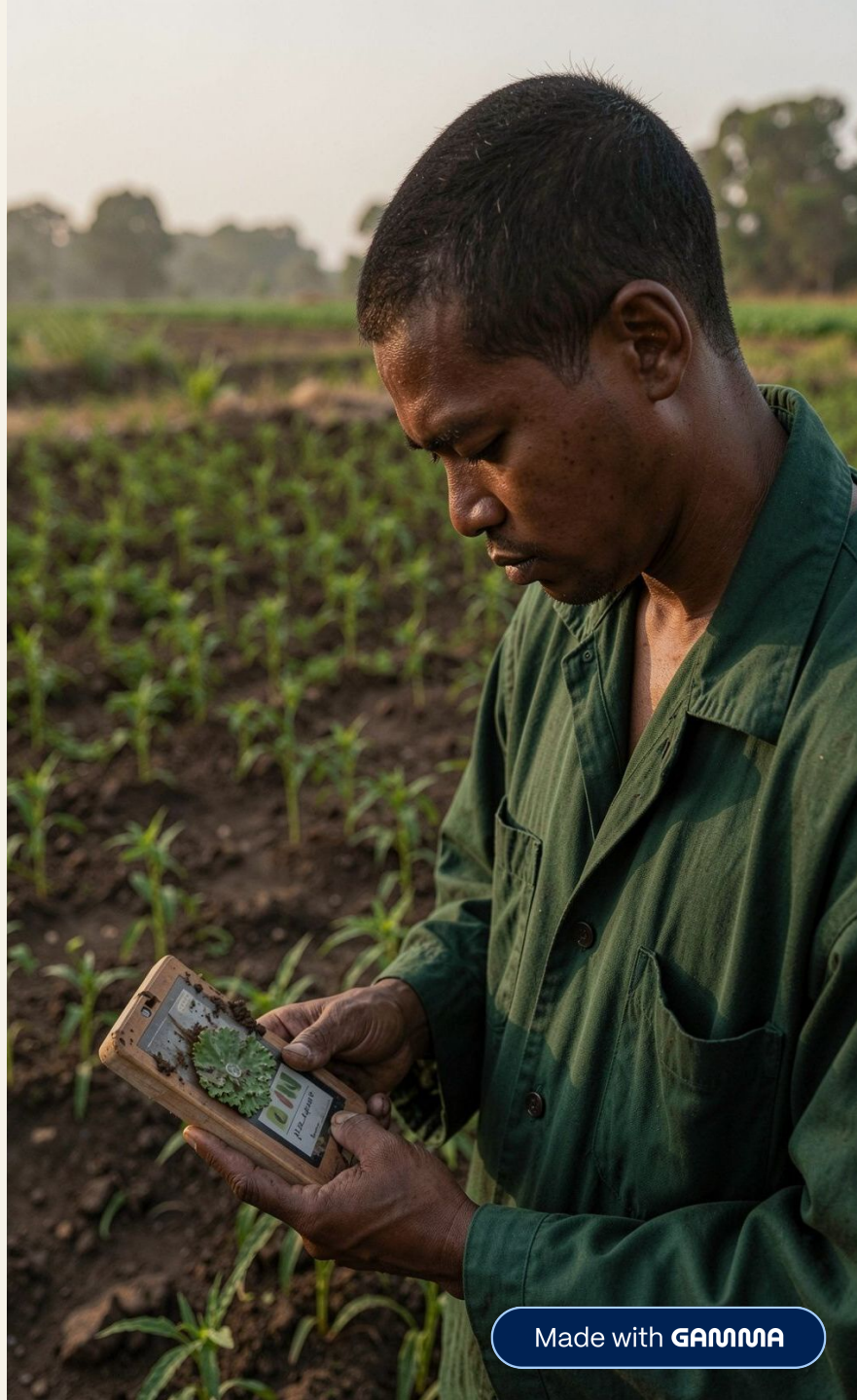
Enable more informed irrigation decisions.

Care

Support more informed crop-care decisions.

Build

Lay a foundation for data-driven agriculture.



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08 · FUTURE SCOPE

From prototype to platform.

THE CURRENT CONCEPT ESTABLISHES THE SENSING AND INTERPRETATION LOOP. THESE EXTENSIONS ARE FUTURE SCOPE.

CURRENT CONCEPT → FUTURE SCOPE → FUTURE PLATFORM



CURRENT CONCEPT

Sensing + interpretation loop

Soil readings become understandable information.



01 · FUTURE

Crop-specific ranges

Reference ranges and recommendations.



02 · FUTURE

Weather integration

Bring weather context into decisions.



03 · FUTURE

Historical tracking

Build a timeline of soil data.



04 · FUTURE

Dashboard evolution

Improve notifications and farmer workflows.



FUTURE PLATFORM

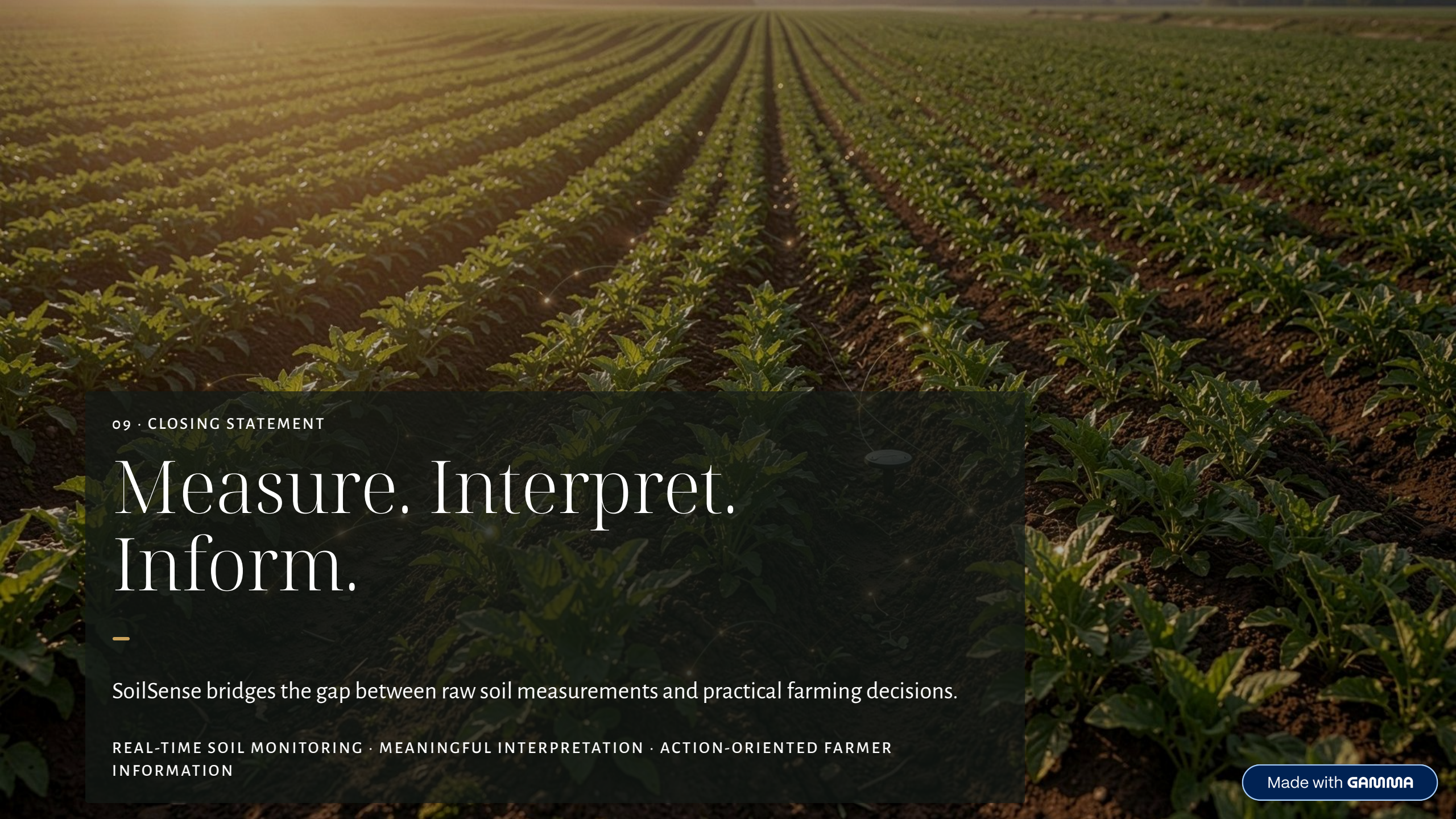
Connected decision support

Extensions remain future scope.

FUTURE ENABLERS

CLOUD CONNECTIVITY · MOBILE APPLICATION · AI/ML RECOMMENDATIONS · PREDICTIVE CROP-CARE

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09 · CLOSING STATEMENT

Measure. Interpret. Inform.

SoilSense bridges the gap between raw soil measurements and practical farming decisions.

REAL-TIME SOIL MONITORING · MEANINGFUL INTERPRETATION · ACTION-ORIENTED FARMER INFORMATION

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